

FROM OPERATIONAL FRICTION TO DIGITAL WORKFLOW.

How a mobile maintenance workflow brought data capture closer to the point of work.

SIM MOBILE

Business Analysis • Operational Excellence • Product Delivery • Change Management

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The problem was not the dashboard. It was the moment data was captured.

Industrial maintenance technicians worked on the production floor, but critical system activities still depended on desktop computers located in support rooms.

Technicians left machines to find an available computer, check stock, open or close work orders and record labor time. When access was difficult, information was often postponed until the end of the shift.

Operational friction and data-quality friction were the same problem.

MY ROLE

Business-side project lead

Translated maintenance workflows into functional requirements, coordinated with IT, validated the solution, supported pilot testing, trained the PCM team and led business-side rollout activities.

2021

initial rollout year

5+

mobile workflows

148

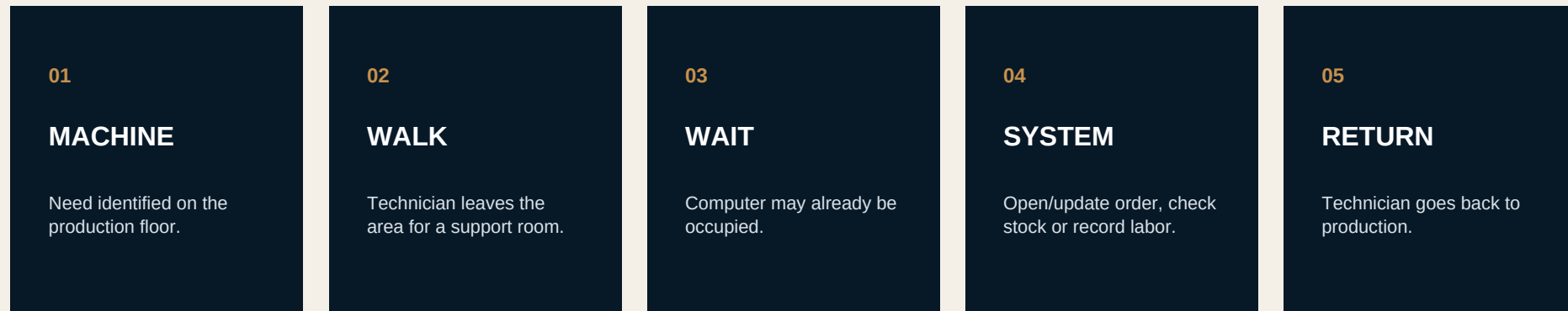
enabled users by Oct/2022

3

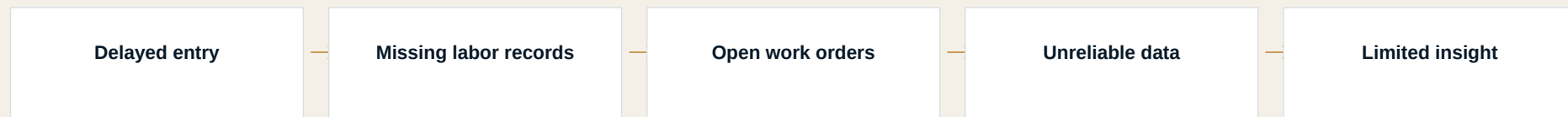
Brazilian units tracked

A technician could fix the machine faster than the process around the repair

The maintenance workflow was physically disconnected from the point of work.



THE HIDDEN DATA CHAIN



“The operation was happening, but the system did not always reflect operational reality.”

Connecting frontline reality with technology delivery.

The idea originated with maintenance management. My responsibility was to execute the initiative from the business side and make the solution work in the real maintenance environment.

I DID NOT BUILD THE SOFTWARE.

I translated operational workflows into functional expectations, validated the product and coordinated business-side implementation.

1 OPERATIONAL PAIN

Hands-on knowledge from maintenance supervision.

2 REQUIREMENTS

Define screens, flows and expected mobile behavior.

3 BUSINESS ↔ IT

Align needs, follow development and clarify workflows.

4 TEST & ITERATE

Functional tests, bugs, corrections and retesting.

5 ROLLOUT

Pilot, training, communication and adoption support.

Start with the highest-friction workflows. Expand after validation.

The solution evolved incrementally instead of attempting a full mobile replica of the desktop system.



DESIGN PRINCIPLE

“What does a technician actually need to do while standing next to a machine?”

The mobile experience was designed around operational moments: opening or closing a work order, checking material availability, recording labor and completing preventive activities closer to where the work occurred.

A technically functional solution creates no value if users do not adopt it.

PILOT LOOP

Develop → Test → Observe

→ Report issues → Correct → Retest

The first pilot used electrical maintenance teams and a limited functional scope. Bugs and workflow issues were identified in real use, corrected with IT and retested before broader expansion.

ROLLOUT MODEL

1. Authorization
2. Training content
3. IT Wi-Fi access request
4. System access
5. Start mobile usage

The PCM team was trained first and then enabled to support maintenance teams.

EFFICIENCY

TIME SAVINGS

DATA RELIABILITY

LESS PAPER

FUTURE USE CASES

Evidence, without overclaiming.

93

enabled users in an earlier project view

148

enabled users by Oct/2022

23%

share in the project population view

WHAT STRUCTURALLY CHANGED

- Key system activities moved closer to the point of work.
- Data could be recorded closer to the activity.
- Less dependency on support-room desktops.
- Stronger foundation for timely operational data.

ANALYTICAL LIMITATION

Access is not the same as active usage.

Historical reports primarily track access enablement and adoption indicators. They do not prove daily active usage, feature utilization or quantified productivity gain.

I therefore do not attribute unsupported financial or productivity results to the project.

The next phase would turn rollout data into product and operational intelligence

ADOPTION

- Monthly Active Users
- Access-to-Usage Conversion
- Feature Adoption Rate

EFFICIENCY

- Time to work-order closure
- Same-shift closure rate
- Mobile vs. desktop share

DATA QUALITY

- Orders without labor allocation
- Open-order aging
- Late data-entry rate

BUSINESS

- Technician productive time
- Maintenance backlog
- MTTR correlation

ANALYTICS OPPORTUNITY

Connect access logs, work-order events and team-level performance to understand who adopted the product, which workflows created value and where operational barriers remained.

What this project taught me about transformation.

01

Digital transformation starts with operational friction.

The starting point was not technology. It was a technician walking away from a machine to find a computer.

02

Data quality often begins before the database.

The problem was not initially a dashboard problem. It was a data-capture problem.

03

Business and technology need a translator.

Maintenance understood the pain. IT understood the technology. The project needed both realities connected.

04

Delivery does not end at launch.

Testing, training, communication, access management and adoption tracking were part of the product journey.

FROM DELAYED ADMINISTRATIVE RECORDS TO DATA CAPTURE CLOSER TO OPERATIONAL REALITY.

Better information creates the foundation for better decisions.